

AIQG — Strengthening the Position with SLM/LLM/ML in the Loop

Status: Working strategy — 2026-06-19. **Not legal advice.** **Family:** [AIQG-PATENT-ANALYSIS.md](#), [AIQG-PROVISIONAL-BRIEF.md](#), [AIQG-FTO-CLAIM-READ.md](#), [AIQG-PATENT-SCOUT-AUTOLEARNING.md](#)

The governing principle

Adding a model to a system is obvious; “use ML to route / score / detect” is prior art many times over. A claim is strong only when the model is anchored to a **signal the gateway uniquely has and competitors structurally cannot get**. The gateway sits where four privileged signals converge:

- **P1 — Forced echo:** the stateless chat protocol guarantees served output (and vendor-minted `tool_call_ids`) reappears in later requests → **free, deterministic ground-truth labels** for flow/parentage.
- **P2 — Inferred per-cluster identity:** the fingerprint-ensemble ladder groups un-instrumented traffic into agent clusters → a **scoping key** no instrumentation-dependent competitor can apply to un-tagged callers.
- **P3 — CLEAR decomposition:** per-response Cost/Latency/Efficacy/Assurance/ Reliability → a **structured supervised target** richer than a thumbs-up.
- **P4 — Closed feedback + shadow execution:** human/implicit feedback plus the already-shipped shadow-extraction pipeline → a **measured-outcome training stream** with realized (not projected) deltas.

Every ML claim below is a model trained on, or scoped by, one of P1-P4. That is the non-obvious, hard-to-replicate part — not the model architecture.

Ranked ML-in-the-loop candidates

M1 — Self-labeled agent classifier (weak supervision from the deterministic tier) □ strongest

Claim core. Train a model to classify/cluster **un-instrumented** traffic into agent identities, using **labels generated by the gateway itself**: the deterministic linkage tier (P1 — `tool_call_id` echo, prefix chains) supplies ground-truth flow boundaries and parentage, and cooperatively-tagged traffic (TAS-Agent-* headers) supplies ground-truth agent names. The model learns to reproduce those labels on traffic that has neither — i.e. **the protocol’s forced echo and the opt-in tags become a free, continuously-refreshed training set; no human labeling**.

Why it’s strong. Narrow, non-obvious, and uniquely enabled by the gateway (P1+P2). It is the ML realization of “tags teach the inferrer,” made concrete: the differentiator is the *label source*, not the classifier. A competitor without the deterministic echo tier cannot manufacture these labels.

Build for reduction to practice: run `cmd/demo-traffic --untagged`, derive labels from the deterministic tier, train a small classifier, report precision/recall per tier vs known ground truth.

M2 — Distilled per-cluster SLM judge, feedback-calibrated (upgrades A2) □ strong

Claim core. Replace/augment the expensive LLM-judge that computes CLEAR **Efficacy/Assurance** with a **small distilled judge model** that is continuously calibrated **per inferred agent cluster (P2)** using the human/ implicit feedback stream (P4) as the supervised target — driving the judge-calibration metrics (agreement/bias/MAE) the dashboard already computes **back into the scorer** as live correction, rather than displaying them.

Why it’s strong. Two differentiated axes at once: (a) **cost** — distilling an LLM-judge into a per-tenant SLM is a measurable efficiency claim; (b) **accuracy** — *per-cluster* online calibration corrects a judge that is systematically biased on one agent type without touching others. Generic LLM-as-judge and RLHF are prior art; **per-inferred-cluster distillation + feedback recalibration scoped by P2** is the combination. Reuses already-shipped feedback + calibration data.

M3 — Predictive CLEAR/cost router keyed on fingerprint features (narrow form of A1)

Claim core. Before routing, a learned model predicts the **decomposed CLEAR vector + cost + latency (P3)** for each candidate model **conditioned on the request’s fingerprint-ensemble features and inferred cluster (P2)**, and routes to the predicted-Pareto-best under a per-cluster guardrail; the prediction is trained on the gateway’s historical (request features → realized CLEAR) pairs and corrected by the live verdict/feedback loop (P4).

Why it needs care. Adaptive/bandit model routing is a **crowded field** (RouteLLM, bandit model selection). File only as the narrow combination: prediction of the **decomposed CLEAR vector** (not a scalar reward) **keyed on inferred-cluster fingerprint features** for **un-instrumented** callers. Run the mandatory bandit-routing prior-art search before claiming.

M4 — Semantic content-propagation (ML upgrade to provisional Candidate 1)

Claim core. Augment the Rabin-fingerprint content-propagation graph with a **small embedding model** so a data-flow edge is detected even when a sub-agent **paraphrases, summarizes, or reformats** the parent’s output rather than copying it verbatim — semantic propagation, IDF/saliency-weighted to suppress boilerplate, scoped per (tenant, principal).

Why it’s useful. Defends Candidate 1 against the obvious design-around (“just paraphrase the handoff and the byte-cache misses”). The novelty remains **zero-cooperation data-flow attribution at an LLM gateway**; the embedding model only widens recall. The attribution doc already anticipates an async embedding cold-path assist — this names it as a claim.

M5 — Learned redaction/gatekeeper tuned to measured CLEAR impact

Claim core. An SLM-based PII/risk redactor whose **aggressiveness is tuned by the measured CLEAR-quality impact (P3) of its own rewrites**, proven via the Gatekeeper-impact shadow experiment — learning the **minimal redaction that preserves answer quality** per cluster, instead of a fixed policy.

Why it’s useful. Turns the security layer into an adaptive component with a quantified quality contract. Ties to existing Gatekeeper/Hyperscan work and the shadow-eval experiment design. Mind FTO around PII-detection patents (separate search).

M6 — Shadow-execution as a learning pipeline (already half-built) □ fastest RTP

Claim core. The gateway runs **candidate SLM transformations** (e.g. the shipped Ollama-embedding + SLM payload-reduction extraction) in **shadow against live traffic**, measures the **realized** cost/quality delta with CLEAR (P3/P4), and **learns which transformation to apply per inferred cluster** from those measured deltas — closing *shadow* → *measure* → *select*.

Why it matters most for filing. Plan #7 Phase 2 already ships the shadow SLM-extraction + measured-vs-projected reduction on the AIQG gateway. That is a **working ML-in-the-loop pipeline today** — the cheapest path to a reduction-to-practice exhibit. The patentable step is adding the **per-cluster learned selection** on top of the existing measurement.

What to avoid claiming (obvious / crowded)

- “Use an LLM/SLM to classify, score, route, or detect anomalies” — bare form is prior art. Always bind to P1-P4.
- “Fine-tune a model to pick the cheapest model” — bandit/RouteLLM territory.
- “Use embeddings for similarity” / “RAG” — generic.
- Any claim where the novelty is the **model**, not the **training signal or scoping**. The model is replaceable; the privileged signal is the moat.

Recommended sequencing

1. **M1 + M6 first** — both have the clearest reduction-to-practice path (M1 via --untagged self-labeling; M6 already partly shipped). Strong, narrow, uniquely gateway-enabled. Fold into the **first provisional** with attribution Candidates 1-4 and Scout S1.
2. **M2 next** — highest-value autolearning claim; prototype the feedback → per-cluster SLM-judge calibration loop, then file in the **second provisional** with the A-series.
3. **M3 / M5** — only after their dedicated prior-art/FTO searches (bandit routing; PII-detection). M4 rides along as a dependent on Candidate 1.
4. **Disclosure hygiene** unchanged: keep designs non-public until the relevant provisional is filed.